

FASTTool v2.0 – User's guide

Michiel Zaaijer

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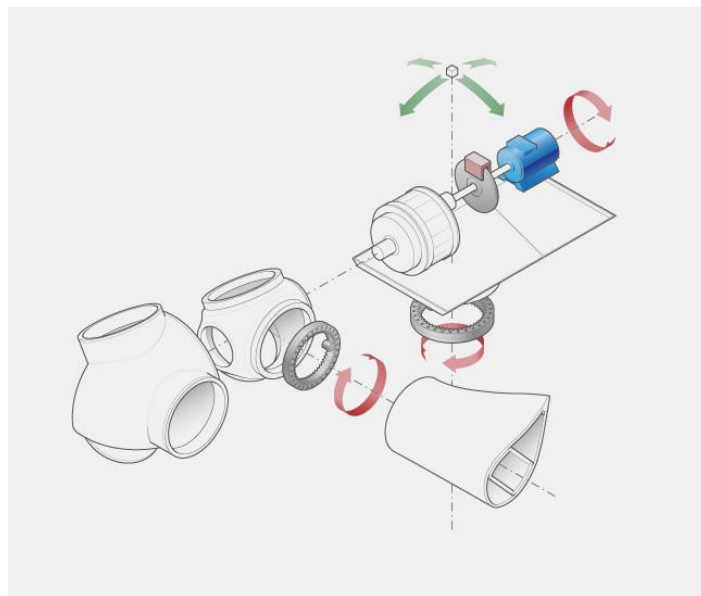


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1 Installing and executing FASTTool

1.1 Requirements

The table below shows the requirements for installing and using FASTTool. Some functions have special requirements. The most important limitation is that the tools from NLR (formerly NREL) that are embedded in FASTTool are only compatible with Windows. Therefore, modal analysis, linearisation and simulation can only be done on Windows.

Item	Function	Operating system			Matlab 2024b (or newer)		Simulink	Control System Toolbox ³
		Windows	MacOS	Unix ¹	Local	online ²		
<i>Menu</i>								
Project	[all]	✓	✓	✓	✓	✓		
Simulink	[all]	✓	✓	✓	✓	✓	needed	
Help	[all]	✓	✓	✓	✓	✓		
<i>Turbine specifications tabs</i>								
Rotor	[edit data]	✓	✓	✓	✓	✓		
Aerofoils	[edit data]	✓	✓	✓	✓	✓		
Tower	[edit data]	✓	✓	✓	✓	✓		
Drivetrain & nacelle	[edit data]	✓	✓	✓	✓	✓		
Control 1	[edit data]	✓	✓	✓	✓	✓		
Control 2	[edit data]	✓	✓	✓	✓	✓		
<i>Analysis tabs</i>								
3D	[all]	✓	✓	✓	✓	✓		
Visual	[all]	✓	✓	✓	✓	✓		
Rotor	[all]	✓	✓	✓	✓	✓		
Steady	[all]	✓	✓	✓	✓	✓		
Modal	[all]	✓	✗	✗	✓	✗		
Control (linear)	Linearise	✓	✗	✗	✓	✗	needed	
	Open	✓	✓	✓	✓	✓		
	Analyse ⁴	✓	✓	✓	✓	✓		needed
Simulation	[edit settings]	✓	✓	✓	✓	✓		
	Run	✓	✗	✗	✓	✗	needed	
Quick scan	[all]	✓	✓	✓	✓	✓		

¹ Not tested

² Not tested extensively

³ When installing FASTTool as add-on, Matlab should automatically detect this dependency, and install the Control System Toolbox if needed. If FASTTool is installed as script files, the availability of the Control System Toolbox must be checked and (if needed) installed manually. Use the button 'Add-Ons' on the Home menu of Matlab to do this.

⁴ Done automatically after 'Linearise' or 'Open' are completed

1.2 Install and run as add-on (recommended)

Notes in advance

- When using a local installation of Matlab, FASTTool must be installed on a local drive (not on a network drive or in the cloud). To check where Matlab installs add-ons, go to the HOME tab in the ribbon and then to 'Preferences' (in the ENVIRONMENT group). The directory where add-ons are installed can be found under 'Matlab > Add-Ons'.
- To install FASTTool in Matlab online, drag and drop the file <FASTTool.mlappinstall> into its 'Files' panel of Matlab online and follow the instructions below. Project files can likewise be dragged and dropped into the 'Files' panel.

To install

- In the Matlab folder window, go to the directory with the file <FASTTool.mlappinstall>.
- Double click the file.
- Click 'Install' in the dialogue box.
- The app has now been added to the Matlab apps under 'MY APPS'. This can be found in the Matlab ribbon in the tab APPS. More apps will be shown by clicking the down-arrow to the right of the box with app icons.
- Click the star on the top-right of the FASTTool icon to add it to your favourites. The star appears when you hover over the icon.

To run

- Make the directory from which you want to run FASTTool the working directory. This would normally be the directory in or below which you want to store the project and results files.
 - When a modal analysis (on tab 'Modal'), a linearisation (on tab 'Control (linear)') or a simulation (on tab 'Simulation') is planned, the chosen working directory may not have a path with more than 85 characters.
 - The working directory may be changed while FASTTool is running. Changes in working directory affect where FASTTool searches for and saves files. Do not change to a working directory that contains a script file with the name <FASTTool>, because this can result in errors.
- In the Matlab ribbon, go to the tab APPS.
- Single click on the icon of the app. If it is not visible in the box of app icons, click the down-arrow to the right of the box.
- Starting FASTTool takes some time, especially the first time of a Matlab session.

1.3 Install and run as script files

To install

- Rename the file <FASTTool.mlappinstall> to <FASTTool.zip> (or add the extension '.zip').
- Create the directory where you want to store the app files (e.g. a directory 'FASTTool').
- Open the file <FASTTool.zip> and copy the entire content of the directory <fsroot> to this directory.

To run

- Make the directory that contains the scripts the working directory.
- In the command window, run the command 'addpath(genpath(pwd))'.
- Make the directory from which you want to run FASTTool the working directory. This would normally be the directory in or below which you want to store the project and results files.
 - When a modal analysis (on tab 'Modal'), a linearisation (on tab 'Control (linear)') or a simulation (on tab 'Simulation') is planned, the chosen working directory may not have a path with more than 85 characters.
 - The working directory may be changed while FASTTool is running. Changes in working directory affect where FASTTool searches for and saves files. Do not change to a working directory that contains a script file with the name <FASTTool>, because this can result in errors.
- In the command window, run the command 'FASTTool' (this is case sensitive – do not run the tool by dragging <FASTTool.mlapp> from the folder window to the command window).
- Starting FASTTool takes some time, especially the first time of a Matlab session.
- The command 'addpath(genpath(pwd))' must be run with the script directory as working directory each time you restart Matlab. If you make the scripts directory a subdirectory of the folder from which you run FASTTool, you can keep that folder as working directory permanently, without needing to switch each time you restart Matlab. Running the command 'addpath(genpath(pwd))' (from that working directory) adds all subdirectories to the Matlab search path.

2 Overview of layout and editing

2.1 Layout

Menu – Actions grouped here

- Managing project files, such as opening, saving, closing and comparing two projects side-by-side.
- Viewing stored results.
- Viewing the Simulink model.

Left panel – Actions grouped here

- Editing data that specify the turbine design and control

Right panel – Actions grouped here

- Performing analysis of the specified turbine, including checking input data in visual form.

2.2 Editing data

Editing cells

- When editing cells, the validity of the data type is checked.
- The fulfilment of most basic requirements on range of validity is checked (such as being positive).
- Consistency (such as increasing heights in table of tower dimensions) is not checked.
- Sensibility (such as the magnitude of the value of stiffness of a blade element) is not checked.

Editing tables

- Individual cell content can be changed in the table.
- Tables can be exported to Excel, for easier editing (including in-table calculations).
- The number of rows in a table can only be changed in exported Excel files.
- Instructions for working with exported data are given in the Excel file.
- When importing Excel files, the entire current table contents is replaced.
- The same checks are performed as when editing cells. When importing tables, a check is performed to establish that the Excel file contains a corresponding table data.
- Table-data inputs can be viewed in the right panel, on the tab 'Visual'. That tab has a checkbox 'Best match with left tab'. When ticked, the visualisation will follow the selection of input tab on the left panel, and will show the most relevant graph.

Restoring data

- Pressing the 'Restore button' retrieves data from the last-saved project file.
- This only affects the data on the active tab and it will overwrite all edits (cells, toggles, drop-down boxes, tables, ...).

3 Actions from the menu

3.1 Working with project files

- Project files contain all the data of the left panel. For the right panel, they contain the 'Siting' data from the tab '3D' and the settings from the tab 'Simulation'. Other settings on the right panel are not stored.
- After opening a project, the data-input and analyses tabs will be shown, with data from the project file.
- Projects can be saved under the same filename or with a different filename from the menu, or saved with the same filename using ctrl-s (lowercase 's', unlike what the menu says) from anywhere in the app.
- An asterisk (*) in the tab title indicates that any project data on that tab has been changed since the last save. The asterisk disappears when the data on the tab is restored.
- Project files do not have to be in the Matlab working directory.
- Two project files can be compared side-by-side. This is done in a separate window. Choosing this option when a project is open will not close the project.

3.2 Viewing results

'Review results' item under 'Project'

- This enables loading results from file and viewing them in the same way as directly after an analysis.
- If there is an open project, it will be closed.
- The analysis panel of FASTTool will be opened, showing only content in tabs for which results can be stored and loaded ('Rotor', 'Steady Op', 'Modal', 'Control (linear)' and 'Quick scan').
- Simulation results can be displayed on the tab 'Quick scan', rather than on the tab 'Simulation'.
- The results viewer can be closed from the menu, or it will close when a project is opened.

3.3 Viewing Simulink model

Items under 'Simulink'

- These open Simulink (in a separate window) and show the chosen model.
- It takes some time to open Simulink, especially the first time in a Matlab session
- The torque, pitch and fore-aft damper controllers are parts of the full model. Once opened, it is possible to move into and out of parts of the full model in Simulink.
- Reviewing the Simulink model helps understanding the setup of the controllers.

4 Turbine data

A general note on turbine data: This user's guide doesn't describe all input parameters. Consult the FAST User's guide for the interpretation of parameters that are used by FAST. Some parameters, particularly for the controller, are input for the Simulink model and not for FAST. These do not appear in the FAST User's guide (e.g. 'Fine pitch angle' in 'Control design').

4.1 Rotor

- The blade ends at the radius of the last entry in the table.
- The number next to the blade root is the number of blades.
- The angle of the blade is the cone angle. The tilt angle of the rotor is input in 'Drivetrain & nacelle'.
- There is no input for pre-bending. Pre-bending can be approximated by increasing the cone angle with a representative value.
- The cell for 'Aerofoil' in the table only allows the selection of aerofoils that exist in the database. To add a new aerofoil, first add it on the tab 'Aerofoils'.
- If a table is imported with an aerofoil name that doesn't exist in the database, an aerofoil with this name will be added to the database, using data of a dummy aerofoil (the same as when adding a new aerofoil on the 'Aerofoil' tab).

4.2 Aerofoils

- Aerofoil coordinates are only used for visualisation (on the tab '3D' on the right panel).
- New aerofoils can be added. All 'Static properties' and 'Dynamic stall properties' must be input to get proper results from simulations.
- There is a selection of base aerofoils that cannot be deleted.
- Aerofoils that are in use in the table on the 'Rotor' tab cannot be deleted.

4.3 Tower

- The tower ends at the height of the last entry in the table.
- The value on the left of the tower drawing is the hub height. The consistency of this hub height, the tower height and the nacelle height is not checked.
- The 'Tower effective density' may be more than the (steel) material density to represent additional internal structures of the tower (flanges, platforms, ladders, etc.). 8500 kg/m³ is used for the reference turbine.
- Several cross-sectional properties are calculated by FASTTool, based on these geometric inputs (e.g. stiffness).

4.4 Drivetrain & nacelle

- Most parameters of the dimensions of the nacelle are only used for visualisation of the turbine. However, the distance from tower centre to hub and the height of the nacelle are used for the simulation model.
- The inertia of the generator is not used for the 'Steady Op' analysis. It is (only) used in simulations.
- There is no separate input for converter efficiency. Include the converter efficiency in the generator efficiency to get consistent power performance.

4.5 Control 1: Supervisory, partial load and actuator limits

- This tab contains all control parameters besides the primary parameters for dynamic full-load pitch control. The latter parameters are set on the tab 'Control 2'.
- Several parameters on this tab are needed for the 'Steady Op' analysis, as detailed in Section 5.3.
- The 'Fine-pitch angle' in the 'Partial-load pitch' panel is the blade pitch angle to correct for the convention to make the section twist angle equal to zero at the blade tip. The column 'Twist' in the 'Rotor' tab plus the fine-pitch angle is the section twist angle for partial load operations.
- The rotor performance can be determined on the 'Rotor' analysis tab for different pitch angles to find the optimum fine pitch angle.
- Be aware that 'Optimal mode gain' is expressed in Nm/(rad/s)². The optimal mode gain K is the constant in the feedforward generator torque controller $\tau_{gen} = K\omega_{gen}^2$ and theory on partial-load control provides an analytical expression to derive its value. The torque control law is used for maximum power extraction

in the region between points B and C. An analysis of the achieved tip-speed ratio and power coefficient in the analysis on the 'Steady Op' tab can be used to check that the optimal mode gain is set properly.

- For the meaning of 'Gen. speed'-s in the 'Torque control' panel, choose 'Torque control curve' on the 'Visual' analysis tab. This curve is also shown when the 'Control 1' tab is opened and the option 'Best match with left tab' is ticked on the 'Visual' tab.
- Speed A should be smaller than speed B and speed C should be smaller than speed D, to create a little slope in these respective sections of the control curve (as opposed to having a constant rotational speed between points A and B, respectively C and D). This requirement is a consequence of the implemented feed-forward control in these regions.
- 'Demanded torque' is the torque in point D, corresponding to full load operation. The torque control curve is set for the high speed shaft if a gearbox is present or the main shaft for a direct drive. Generator and converter efficiencies need to be taken into account to set the demanded torque that achieves the desired rated (electrical output) power.
- For proper (supervisory-) control behaviour, operating point D should be (at least slightly) above the parabolic blue-dashed curve that represents optimal operation. If needed, 'Gen. speed ω [D]' must be lowered, with a corresponding increase in 'Demanded torque' to maintain the desired rated power.
- Other parameters on this tab are only needed for linearisation and/or simulation.
- Suggestions for 'Torque' settings in the 'Actuator limits' panel (not needed for steady-performance analyses):
 - Minimum: This is the lowest torque that the controller can set for the generator, due to physical limitations. It should be below the minimum torque required at cut in and can be set e.g. by scaling it with respect to that torque level.
 - Maximum: This is the maximum allowable torque that the generator can handle. It is recommended to maintain a similar (relative) margin to 'Demanded torque' as the reference turbine.
 - Slew rate: This is a control-rate limit option, limiting the ability of the controller to change the torque very rapidly. This prevents rapid oscillations to occur. However, when it is too small the torque will not respond sufficiently fast to changes in rotor speed, due to changes in wind speed. It is recommended to use a value roughly half the 'Demanded torque'. When for example the demanded torque is 1,000,000 Nm, the slew rate is recommended to be set at 500,000 Nm/s. This allows the controller to reach the demanded torque in less than 2 seconds.

4.6 Control 2: Full-load pitching and fore-aft damping

- Most of the parameters on this tab are settings for the pitch controller used in full load. This can be either a controller with constant gains (for all wind speed), or with scheduled gains (different for different wind speeds). Instead of wind speed measurements, the controller uses the pitch angle as an indication of the wind speed, to adjust the gains.
- Setting the gains can be done with loop shaping, which uses information from the control analysis of a linear wind-turbine model on the analysis tab 'Control (linear)'. This is further addressed in Section 5.5.

5 Analyses (running and viewing results)

A general note on analyses: all analyses are performed with the current values of input data, even if these have not yet been saved. Thus, effects of changing inputs can be tested without saving the project file, with the option to restore the data to the original values.

5.1 Visualisation of turbine (3D) and inputs

Data used

- Data from any or all tabs from the left panel are used, depending on the chosen visualisation.

3D visualisation

- The 3D images can be rotated by dragging them with the mouse, to view them from different angles.
- By clicking 'Undock', the image is reproduced in a separate window, with a higher resolution.
- The animation is at the rated rotational speed (point D of the torque-control curve).
- The settings for 'Siting' only affect the visualisation. They have no effect on the analyses of the turbine.

Visual reproduction of input data

- All tabulated data from the tabs on the left panel (including the parameterised torque-control curve from the tab 'Control 1') can be visualised on the tab 'Visual'. This can help spotting mistakes or irregularities in the inputs.
- By ticking 'Best match with left tab', the chosen graphs on this tab follow the selection of the tab on the left. If there are too many parameters on that tab, the most important ones are visualised. In that case, the selected graphs can be changed manually under 'Display'. For aerofoils, the selection of 'Aerofoil geometry' or 'Aerofoil polars' will persist to be the best match until unticking 'Best match with left tab'.

5.2 Rotor performance (including c_p - λ curve)

Data used

- Rotor
 - Blade geometry and properties
 - Radius
 - Chord
 - Twist
 - Aerofoil
 - Rotor geometry
 - number of blades
- Aerofoil
 - Static aerofoil polars (of aerofoils selected on 'Rotor')
 - α
 - c_l
 - c_d
- Control 1
 - Partial-load pitch
 - Fine-pitch angle

Performing the analysis

- Running the analysis on the tab 'Rotor' provides the performance of the rotor as a function of the tip speed ratio.
- The performance is expressed in dimensionless form: Power, thrust and torque coefficients.
- A custom range of pitch angles can be set to find the optimal fine-pitch angle (from a small range) or to analyse rotor performance during pitch-control (over a larger range).

5.3 Steady operation (including power curve)

Data used

- All parameters used for rotor performance (see Section 5.2).
- Drivetrain & nacelle
 - Drive train

- Gearbox efficiency
 - Generator efficiency
 - Transmission ratio
- Control 1
 - Torque control
 - Optimal mode gain
 - Gen. speed ω [A] ... [D]
 - Demanded torque
 - Supervisory control (wind speeds)
 - Cut-in
 - Cut-out
- Simulation
 - Air density

Performing the analysis

- Running the analysis on the tab 'Steady Op' provides the performance of the turbine at different wind speeds.
- The performance is expressed in both absolute values (power, thrust, etc.) and dimensionless form (power coefficient, tip speed ratio, etc.).
- The torque control curve set on the tab 'Control 1' is used to determine the operating conditions at each wind speed. The pitch angle settings above rated wind speed are based on steady-state c_p -lamda curves, and not on the PI controller.
- Besides showing the main performance of the turbine with the power curve, the steady state behaviour of the turbine can be analysed with the thrust and torque curves. The rotational speed and the dimensionless parameters are useful to check that the torque control curve is set properly.

5.4 Modal analysis

Data used

- Rotor
 - Blade geometry and properties
 - Radius
 - Twist
 - Masse density
 - Flap. stiff.
 - Edge. stiff.
 - and Rotor geometry
 - Cone angle
 - Number of blades
- Tower
 - Dimensions
 - Height
 - Diameter
 - Wall thickness
 - Hub height and material
 - Tower effective density
- Drivetrain & nacelle
 - Nacelle
 - Hub mass
 - Nacelle mass
 - Drive train
 - Transmission ratio
- Control 1
 - Torque control
 - Gen. speed ω [A]
 - Gen. speed ω [D]
 - Partial-load pitch
 - Fine-pitch angle

Performing the analysis

- The modal analysis can only be performed on Windows OS.
- Running the analysis on the tab 'Modal' determines the natural frequencies of the support structure and of the rotor. Natural frequencies of the rotor are determined for five rotational speeds, to account for the effect of centrifugal stiffening.
- Excitation frequencies and natural frequencies can be switched on and off in the plot, depending on which are relevant to the analysis.
- Displaying mode shapes is only possible after running the modal analysis, until the project is closed. The mode shapes are not part of the saved results that can be viewed later (from the menu-item 'Review results' under 'Project').

5.5 Turbine-model linearisation and (linear) control analysis

Data used for linearisation

- Rotor
 - all data.
- Aerofoils
 - Dynamic stall
 - all data
 - Static aerofoil polars (of aerofoils selected on 'Rotor')
 - all data
- Tower
 - all data.
- Drivetrain & nacelle
 - Nacelle
 - Hub mass
 - Nacelle mass
 - Tilt angle
 - Offset of hub from tower centreline
 - Drive train
 - Gearbox efficiency
 - Generator efficiency
 - Transmission ratio
 - Generator Moment of inertia
- Control 1
 - all data, **except for**
 - Supervisory control
 - Startup
 - Shutdown
 - Torque control
 - Low-pass cut-off
 - Actuator limits
 - all data
- Simulation
 - Air density

Data used for control analysis (with existing linear model)

- Linear model (from file)
- Control 1
 - Partial-load pitch
 - Fine-pitch angle
- Control 2
 - all data, **except for**
 - Fore-aft damping
 - all data

Linearisation

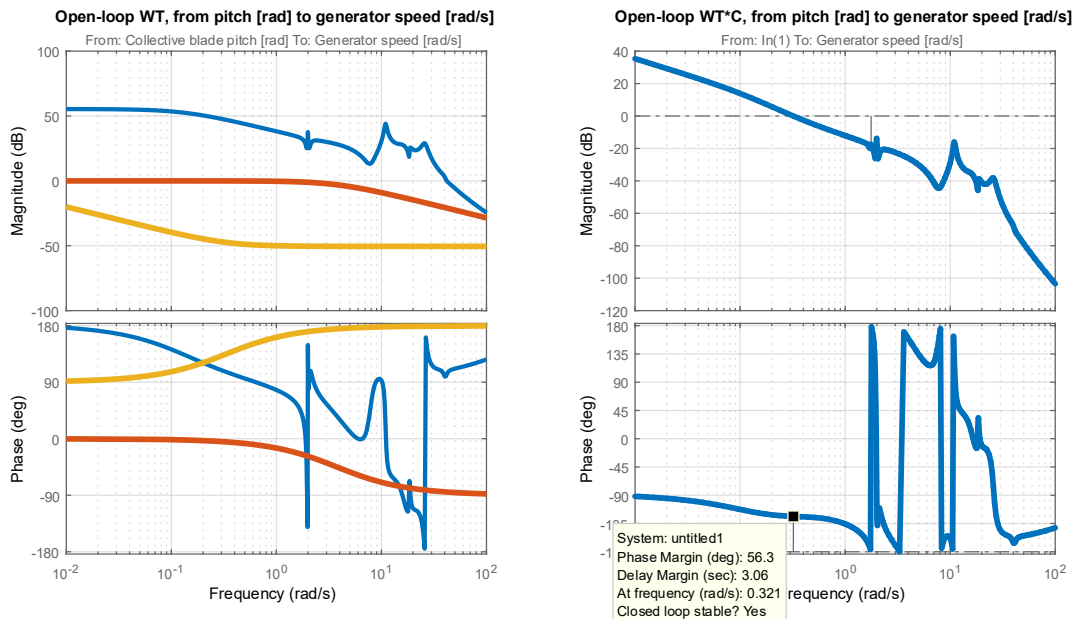
- The linearisation can only be performed on Windows OS.

- The analysis of the controller on the tab 'Control (linear)' uses a linear model of the wind turbine. This model is generated by clicking 'Linearise'. The linearisation is done by running simulations with FAST in Simulink. Be aware that this may take up to around an hour.
- If a linearised model is already available, it can be opened by clicking 'Open' (see 'Control analysis' below).
- The simulations are performed with the sampling time that is set on the tab 'Simulation'. See Section 5.6 for information about the sampling time.
- One linearisation is performed per integer step in wind speeds and only for wind speeds above rated. To test that the linearisation works, or to linearise only for a smaller range of wind speeds, reduce the operational wind-speed range in the 'Supervisory control' panel on the tab 'Control 1'.
- If the linearisation fails for higher wind speeds, the linear model for the earlier steps will still be saved. If the linear model is missing for only a few wind speeds near cut-out, the control analysis and gain scheduling can be performed with the reduced information (see 'Control analysis' below).

Control analysis

- If a linearised model is already available, it can be opened by clicking 'Open'. The consistency of the linear model with the wind-turbine data of the currently open project is not checked.
- The control analysis is immediately performed when opening a file with a linear model, by applying the control settings of the tab 'Control 2' to it. Each time a setting of the controller is changed on 'Control 2', the analysis is immediately redone, and the plots are redrawn.
- The remaining elements on this tab provide the means to choose which of the results to show in the plots:
 - The 'Operating points' provides the pitch angles for which a linear model exists in the opened file. Each pitch angle represents a wind speed, which is shown in square brackets. The controller will use the pitch angle for gain scheduling, rather than the wind speed.
 - The top three 'Plot options' help to understand the turbine response, with (= 'Loop gain') and without (= 'Nominal plant') controller. The option 'Margin' shows for the loop gain what the magnitude of the gain and phase margin are, and at which frequency in the plot they are determined.
 - The bottom three 'Plot options' show the behaviour of the controller elements (a Proportional-Integral (PI) controller, a first/second-order low-pass filter and one or two notch filters). This is their behaviour without being connected to the wind turbine, which helps to see how these elements affect the loop gain (magnitude and phase). On a dB scale for magnitude, the loop gain is the sum of the nominal plant and the control element. The phase of the loop gain is also the sum of the nominal plant and the control element. When only a control element is shown, it can be seen how its effect changes with the settings of the controller. Understanding this helps choosing the control elements and their settings to obtain the desired effect on the loop gain and (therefore) on the margins.
- The pitch angles of the operating points on the tab 'Control (linear)' do not have to be the same as those in the gain-scheduling table on the tab 'Control 2'. However, it is convenient when they are the same, because that facilitates the loop shaping: the tuning of the control settings to get the desired loop gain and margins. Loop shaping can best be done by working on the settings on the left for the same pitch angle for which the control behaviour is shown on the right. By clicking '< Reschedule', a new gain-scheduling table is generated on the tab 'Control 2', with the pitch angles copied from the linear model. The existing settings of the controller will be translated to the new angles by interpolation. If a new angle falls outside the original range of angles in the gain-scheduling table, the settings of the lowest, respectively highest pitch angle will be used for that angle.
- If the linear model is missing for a few pitch angles (thus, wind speeds) near cut-out, the control analysis and gain scheduling can be performed for the available pitch angles and extrapolated to the missing higher wind speeds. Add extrapolated gains to the table on the tab 'Control 2' by exporting, updating and importing the table. The pitch angles corresponding with wind speeds outside the range can be found from the 'Steady Op' analysis. If no gains are provided for higher pitch angles, the controller will use the gains of the highest provided pitch angle in the table.
- A best practice for loop shaping with gain scheduling is to first select 3 wind speeds (e.g. 15, 20, 25 m/s), tune the controller for these, and then tune the controller for the intermediate wind speeds. During these steps, use the three 'Pitch: ...' plots on the tab 'Visual', to graphically represent the gain-scheduling table. A smooth, monotonic trend for all parameters can usually be expected to be preferable.
- An example of loop shaping for the NREL5MW reference wind-turbine controller is given below for a wind speed of 15 m/s. Left plot: the blue line is the open-loop (= nominal plant) Bode plot for the transfer function from pitch angle [rad] to generator speed [rad/s], the orange line is the low-pass filter, and the yellow line is the PI-controller. Right plot: adding up all these components yields the loop gain. From this

loop gain, the gain- and phase margins are evaluated, as well as the cross-over point: the frequency at which the loop gain crosses the 0 dB line. Below the cross-over point, the gain of the closed loop is positive, meaning that the system responds to a disturbance at those frequencies. The higher the cross-over frequency, the faster the controller, but this often comes at a cost for gain- and phase margins (stability). This controller design can be further improved by using a notch filter on the large resonance peak on the right. If that can be suppressed, the margins increase, or the gains can be re-tuned to obtain a higher cross-over frequency.



5.6 Simulations (running and using 'Quick scan' for viewing results)

Data used

- Rotor
 - all data
- Aerofoils
 - Dynamic stall
 - all data
 - Static aerofoil polars (of aerofoils selected on 'Rotor')
 - all data
- Tower
 - all data
- Drivetrain & nacelle
 - Nacelle
 - Hub mass
 - Nacelle mass
 - Tilt angle
 - Offset of hub from tower centreline
 - Drive train
 - Gearbox efficiency
 - Generator efficiency
 - Transmission ratio
 - Generator Moment of inertia
- Control 1
 - all data, possibly except for, depending on chosen Operation
 - Supervisory control
 - Startup
 - Shutdown
- Control 2
 - all data

Preparing and running simulations

- A simulation can only be performed on Windows OS.
- Simulations can be run on the tab 'Simulation', for various wind and operational conditions. Depending on the chosen conditions, different settings become relevant and available.
- Based on the settings, wind fields are generated according to IEC standard 61400-1 ed.3. For wind types EWM1 and EWM50 the average wind speed is set automatically.
- A single simulation can be performed for one (average) wind speed, or a series of simulations can be performed for a range of wind speeds. Simulations can be repeated for different realisations of the wind field for each average wind speed by setting the 'Number of seeds'.
- If a 'First seed nr.' is specified, specific seed numbers will be used for the realisation(s) of wind fields. If 'Number of seeds' is larger than 1, the seed number for simulation 2, etc., will be derived deterministically from this first seed number. Re-running a simulation (or a batch of simulations) with all simulation settings and the first seed number the same will use the same wind field realisation(s). This can be useful when comparing for instance different control settings.
- When multiple wind speeds and/or wind-field realisations (= seeds) are used, the results for each simulation will be saved to a different file. The filenames will be derived automatically from the (single) filename provided after clicking 'Run'.
- When 'Stepped wind' is selected as wind-condition type, only a single simulation will be performed. The stepped-wind simulation is particularly useful to analyse control performance on the non-linear wind-turbine model.
- A period of time can be set for 'Startup [discard]' to allow for the transient response of the turbine during startup of the simulation. This is the result of incorrect estimation of starting conditions (in particular rotational speed and/or pitch angle). The simulation results for this period will be discarded and not appear in the saved results. Typically, this transient behaviour disappears within less than 60 seconds
- The default value for the 'Sampling time' is 0.008 s. Changing this value is usually not necessary, but if the value is smaller, higher-frequency dynamics of the wind turbine is captured. For the NREL5MW reference turbine the default value is sufficient. However, smaller turbines typically have higher natural frequencies and excitation frequencies, which might require a lower value for the sampling time to avoid numerical issues. For larger turbines a larger sampling time could be acceptable.
- The downside of a lower value for the sampling time is an increase in computation time for a simulation (or linearisation).
- At the start of the simulation(s) a window with a scope will appear, from which a few main parameters can be followed while the simulation progresses. The scope has a button 'Scale X & Y Axes Limits', represented by an icon with arrows in top-down-left-right directions. Clicking this button will change the range of the axes to include all currently plotted data points.

Viewing (with 'Quick scan') results

- Results from the simulation are saved, but not shown directly after the simulation. They can be loaded and viewed on the tab 'Quick scan'.
- When loading results on the tab 'Quick scan', the parameter of interest can be chosen from a list and will be shown as a time series and as a spectrum plot. Depending on the type of simulation (continuous conditions, or transient behaviour), either of these will be most relevant.
- These plots are primarily intended to give a first impression. It can be useful to browse through some parameters to get a better understanding of what is happening with the turbine.
- The parameters in the list of results use the naming provided and described in the FAST User's guide. When the results are loaded to the Matlab workspace, a short description can be found by double clicking the variable 'Info' in the workspace, and then double clicking 'dataDescription' in the table that opens for this variable.
- By loading saved results to the Matlab workspace, more analysis, viewing and post-processing can be done. Matlab scripts can be used for post-processing, or data can be converted to be processed in another environment.

5.7 Working with stored results and simulation results

- Results that have been saved for any type of analysis can be loaded to the Matlab workspace for further analysis, viewing and post-processing. They can also be viewed from within FASTTool, as described in Section 3.2.
- Each results file contains the variable 'Info'. Double clicking this variable in the workspace will open a table with the variables 'creationTime', 'dataDescription', 'projectFile' and 'FASTToolVersion'. The results file of a control analysis will contain the variable 'linearModelFile' in addition. 'dataDescription' gives a

brief description of the variables that have been saved. The other variables help tracking to which inputs and FASTTool version the results relate.

6 Comparing projects

Comparing and merging project files

- Open the comparison window from menu-item 'Project > Compare'.
- While this window is open, the main window is not accessible. Nevertheless, an open project in the main window remains open.
- The comparison works with saved project files. It cannot be used to compare a file with changes in an open project.
- Use the 'Browse' buttons to choose the project files to compare. The order in which files are chosen doesn't matter.
- After choosing files, the left panel shows which data groups differ (if any) with an asterisk (*). Selecting the data group in this panel, shows the respective data of the two files. Aerofoil data is not shown, because it is too extensive.
- To merge data from the two files, select under 'Selection' which data to copy to the merged file. The option to select is only relevant and available when the data differs between the two files.
- After selection, the merged file can be saved using the 'Save as ...' button.
- Click 'Cancel' to leave without creating a new file.